

Explainable Human Motion Computing with KSAS and XROCKET

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Abstract

This report applies XROCKET to the course-provided KSAS smartphone inertial dataset to study American Kenpo Karate Blocking Set I. The goal is not only to classify movement labels, but to answer where the information appears in the sensor axes, at what temporal scale it appears, and whether discriminative patterns can be interpreted from a human motion computing perspective. The final evaluation uses participant-grouped folds to avoid train-test participant leakage. The primary XROCKET random-forest model reached mean macro F1 0.883 over five folds; a logistic-regression sensitivity model reached mean macro F1 0.884. The strongest stable explanation evidence points to gravity and device-frame z-axis structure, long temporal spans in the padded 56-sample representation, and representative PPV pattern intervals that are plausible but not sufficient for validated coaching or biomechanical causation claims.

Introduction

Human Motion Computing treats movement as both an action and an information source. In this project, the action is a set of karate blocking movements, and the information source is a smartphone inertial measurement stream. The selected assignment scope is Part I - Human Motion Computing with Inertial Signals. The technical aim is to apply XROCKET, an explainable time-series classifier, to the KSAS dataset and provide evidence for three required questions:

1. Which sensor axes contribute most to movement classification?
2. Which temporal scales are most informative?
3. Which discriminative patterns appear meaningful from a human perspective?

XROCKET was selected because it exposes feature metadata such as channel, dilation, kernel length, and threshold (dida Data Science 2024a, 2024b). This metadata makes it possible to trace model behavior back to sensor-coordinate evidence instead of reporting only aggregate accuracy.

Dataset

The dataset is the course-provided KSAS smartphone IMU dataset (Human Motion Computing Course 2026). The audited local copy contains 240 CSV recordings: 20 participants, two arms, and six movement labels with 40 samples per label. The labels are no movement, upward block, hammering inward block, extended outward block, outward downward block, and rear elbow block. Each CSV contains 18 channels: accelerometer, gravity, gyroscope, linear acceleration, game rotation vector, and magnetic field, each with x, y, and z device-frame axes.

The audit found no missing values, no duplicate rows, no duplicate files, and no schema errors. Sequence lengths vary from 18 to 56 samples, with median length 33. Raw CSVs are kept under `data/raw/KSAS-Dataset/` locally but are ignored by Git and are not redistributed in the repository.

Methods

The repository builds a manifest from the audited raw files, converts the 240 recordings into a tensor with shape (240, 18, 56), and right-pads shorter recordings with zeros. The tensor contract records sample order, channel order, labels, participant IDs, arm metadata, original lengths, and valid-timestep masks.

Evaluation uses five participant-grouped folds. All samples from a participant remain in the same fold, and each fold contains 48 held-out samples with eight samples per class. The reported metrics therefore test participant-independent generalization rather than sample-level interpolation.

Baselines use masked statistical features with majority, logistic-regression, and random-forest classifiers. The primary XROCKET experiment fits encoder thresholds on training folds only, extracts 9,072 PPV features per fold, and trains a random forest. A logistic-regression model on the same transformed features is retained as a sensitivity check.

The XROCKET implementation is pinned to `dida-do/xrocket` commit `1511e810c59d0c42f6431ef2f1f9fa57c71e9b2f` (dida Data Science 2024b). The public upstream repository had no public license file during this project, so this report treats its use as course-authorized academic support and does not claim a general open-source license for the upstream code.

Performance Summary

The statistical random-forest baseline reached mean macro F1 0.909, while the primary XROCKET random forest reached mean macro F1 0.883. This means the selected XROCKET representation is not the strongest classifier in the repository, but it is the main explainable model because it provides traceable channel and dilation metadata.

Model	Mean macro F1	Mean balanced accuracy	Notes
Majority baseline	0.048	0.167	Sanity baseline
Statistical logistic regression	0.886	0.888	Masked statistical features
Statistical random forest	0.909	0.908	Strongest baseline
XROCKET random forest	0.883	0.883	Primary explainable model
XROCKET logistic regression	0.884	0.883	Sensitivity model

The primary XROCKET random forest was weakest for labels 2 and 3, with recalls 0.750 and 0.775. These are the hammering inward block and extended outward block, and their mutual confusion is discussed as a limitation.

Task 1.1 - Sensor-Axis Contribution

Answer to Task 1.1

My answer is that the most useful signal group is gravity, and the strongest pooled axis is the device-frame z axis. Gravity has mean normalized importance 0.200, ahead of gyroscope at 0.175 and accelerometer at 0.168. When I group all sensors by axis, z is highest at 0.347. This does not mean that only one axis matters, but it tells me where the model found the most stable information.

Figure 1 combines the two main pieces of evidence for this answer. On the left, the bar chart ranks the sensor families, where gravity is clearly first. On the right, the heatmap shows how each family splits across the x, y, and z device-frame axes; the brightest gravity cell is on z. I use this panel for the main answer because it shows both “which sensor” and “which axis” in one place.

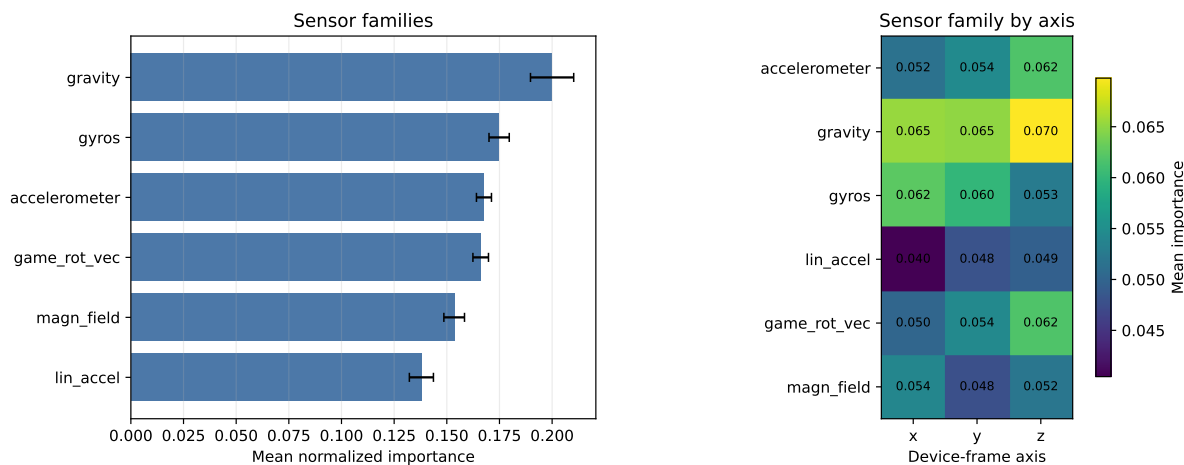


Figure 1: Task 1.1 core evidence. Gravity is the strongest sensor family, and the pooled device-frame z axis carries the strongest axis evidence.

Figure 2 is more detailed and should be read as a secondary diagnostic. It shows that different movement classes emphasize different channels, but I do not use it for the strongest claim because individual channel ranks were less stable than the broader family and axis results.

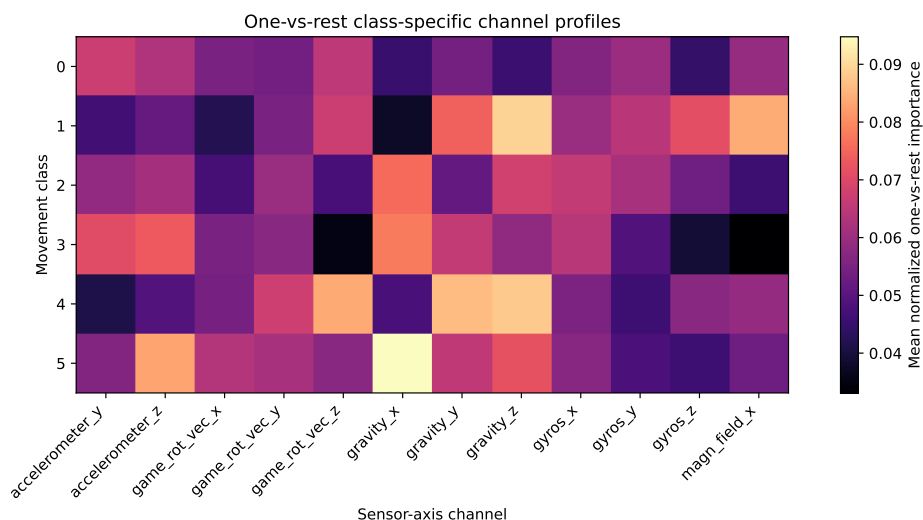


Figure 2: Task 1.1 class-specific channel profiles. These one-vs-rest diagnostics show movement-specific channel emphasis but are secondary evidence.

I checked the ranking against validation tests instead of trusting one score only. The grouped permutation test again points to gravity as the most important sensor family, followed by game rotation vector and gyroscope. The ablation plot is less direct: removing one feature group often causes only a small drop, or even a negative drop. I interpret this as redundancy. XROCKET creates many related transformed features, so the model can sometimes compensate when one group is removed.

Figure 3 puts the validation checks together. The ablation panel is a warning against a simple “one sensor causes the classification” story. The stability panel shows why I trust broad family and axis conclusions more than claims about one exact channel such as **gravity_z**.

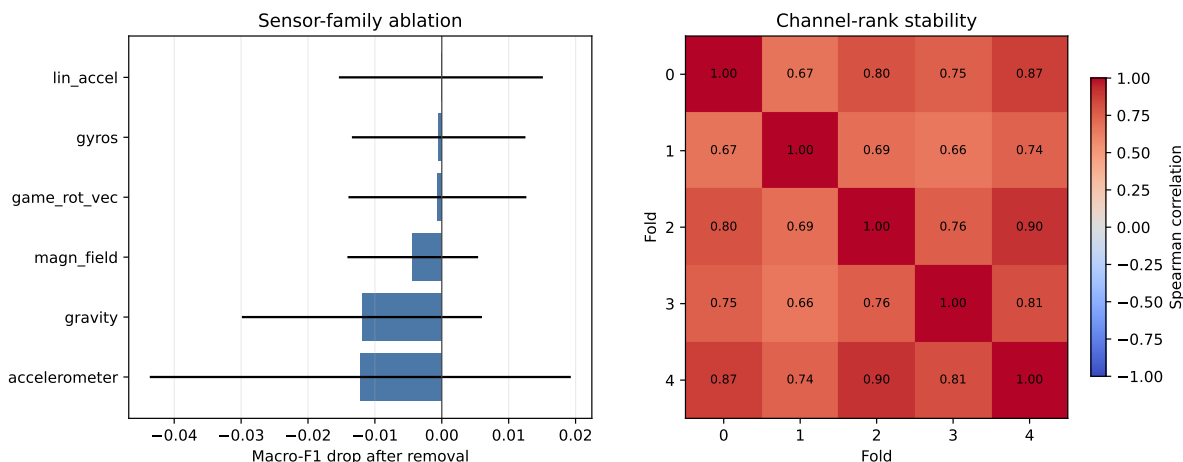


Figure 3: Task 1.1 validation checks. Small ablation drops indicate feature redundancy, while fold stability shows that broad conclusions are safer than individual-channel claims.

My biomechanical interpretation is cautious. The result makes sense because a phone on the forearm should capture changes in orientation, rotation, and acceleration during karate blocks. Gravity and rotation-related channels are therefore plausible sources of movement information. At the same time, these are Android device-frame axes, not anatomical axes. Without calibration to the body, left/right mirroring checks for every movement phase, or expert-labeled biomechanical phases, I should not claim that the model has identified exact joint mechanics, muscle activation, force, or skill quality.

Task 1.2 - Temporal-Scale Analysis

Answer to Task 1.2

My answer is that the model relies mainly on long-duration movement structure, not only on short impulses. With kernel length 9, the XROCKET dilations map to spans of 9, 17, 25, 33, 41, and 49 samples. The long spans, 41 and 49 samples in the padded 56-sample window, account for mean normalized importance 0.655. The intermediate spans account for 0.224, and the short spans account for 0.122. The single strongest dilation is 6, which corresponds to a 49-sample receptive field.

Figure 4 combines the dilation and scale evidence. The left panel shows that dilation 6 is much higher than the shorter dilations. The right panel simplifies the same result into short, intermediate, and long bins: the long-scale bin dominates. This supports the idea that XROCKET is recognizing the overall shape of the block, not just a quick spike at one instant.

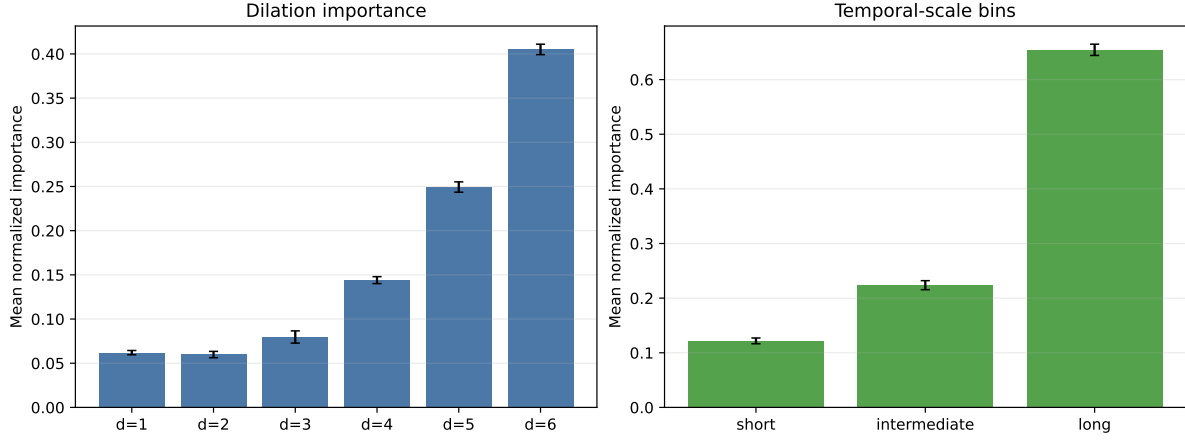


Figure 4: Task 1.2 scale evidence. Dilation 6 and the long temporal-scale bin dominate the saved padded XROCKET representation.

I keep the time interpretation in samples first. The KSAS app requested Android `SENSOR_DELAY_GAME`, which I treat as about 50 Hz only for rough context, but the exported CSV files do not contain timestamps. Android sensor delays are also requests rather than guaranteed exact sampling intervals (Android Developers 2026). For that reason, the strong evidence is “41 and 49 samples”, not a verified number of physical seconds.

Figure 5 adds two checks. The left panel shows that most classes have their highest importance in the long-scale column. The right panel shows that the dilation ranking is stable across folds, so the long-scale conclusion is not coming from one unusual split.

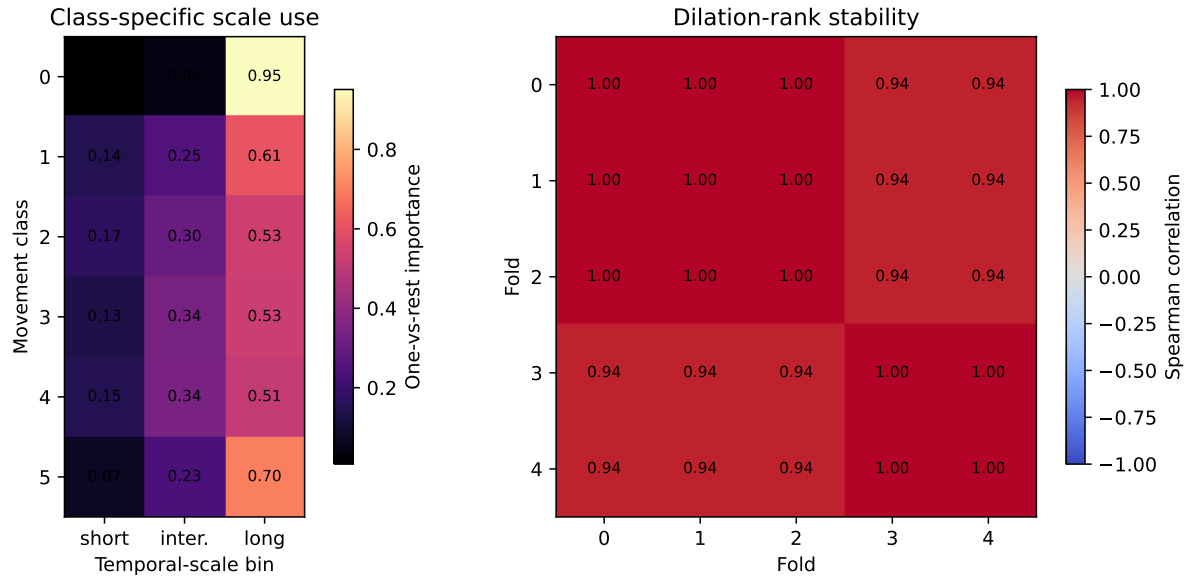


Figure 5: Task 1.2 class and stability checks. Most classes rely most on long spans, and dilation-rank correlations stay high across folds.

My interpretation is that these karate blocks are distinguished by broad arm trajectory and orientation changes. That fits the assignment question because the temporal scale affects what the model can see: short spans would emphasize brief impacts or local changes, while long spans capture a larger part of the movement.

The main caveat is padding. XROCKET does not use the valid-timestep mask, and padding diagnostics showed that padding can affect some predictions and features. A timestamped or mask-aware version would be needed before making stronger claims about real physical timing.

Task 1.3 - Discriminative-Pattern Interpretation

Answer to Task 1.3

My answer is that the selected XROCKET patterns are meaningful as cautious sensor-coordinate explanations. They show broad movement shape, orientation change, and acceleration structure. They do not prove a biomechanical cause, and they are not enough to build a coaching or performance-scoring system by themselves.

The important detail is how PPV features work. A PPV feature is the proportion of convolution responses above a learned threshold. So the highlighted interval in each case is the strongest representative interval for that pattern, not a single exact moment that caused the prediction. Most selected patterns are long-span patterns at dilation 5 or 6. The clearest examples involve `lin_accel_y`, `gravity_z`, and `lin_accel_z`, mainly for upward block and hammering inward block.

Figures 6 and 7 show the four pattern cases as compact cards. Each case card has the raw same-family sensor traces on top, the XROCKET response under it, and a compact metrics strip below. The orange bands show the representative interval used for interpretation. Figure 6 contains the plausible upward-block examples; Figure 7 keeps the ambiguous and failure cases visible so the explanation limits are not hidden.

Task 1.3 plausible representative patterns

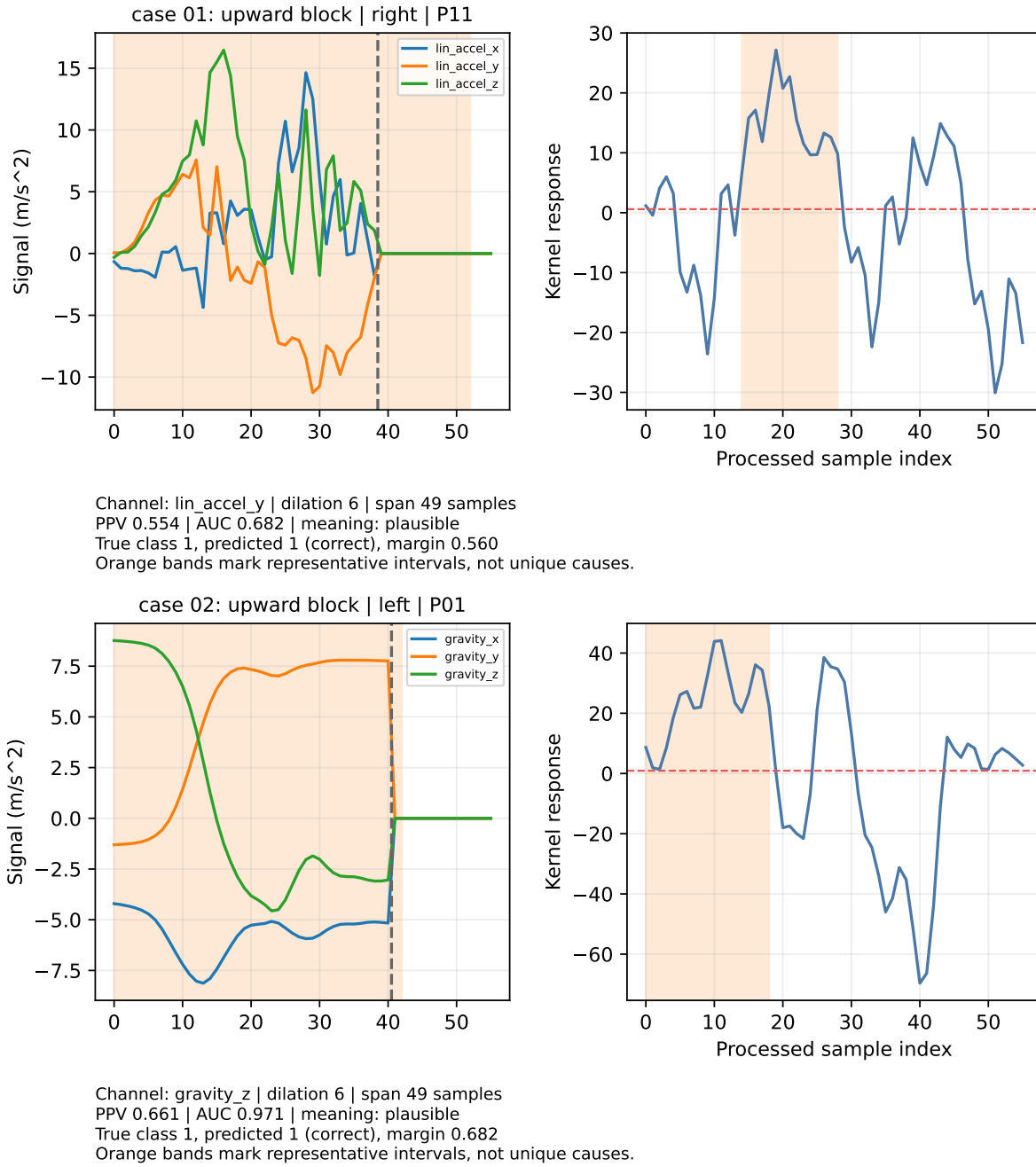


Figure 6: Task 1.3 plausible pattern case cards. Correct upward-block examples show long-span linear acceleration and gravity patterns used by XROCKET.

Task 1.3 ambiguous and failure-pattern checks

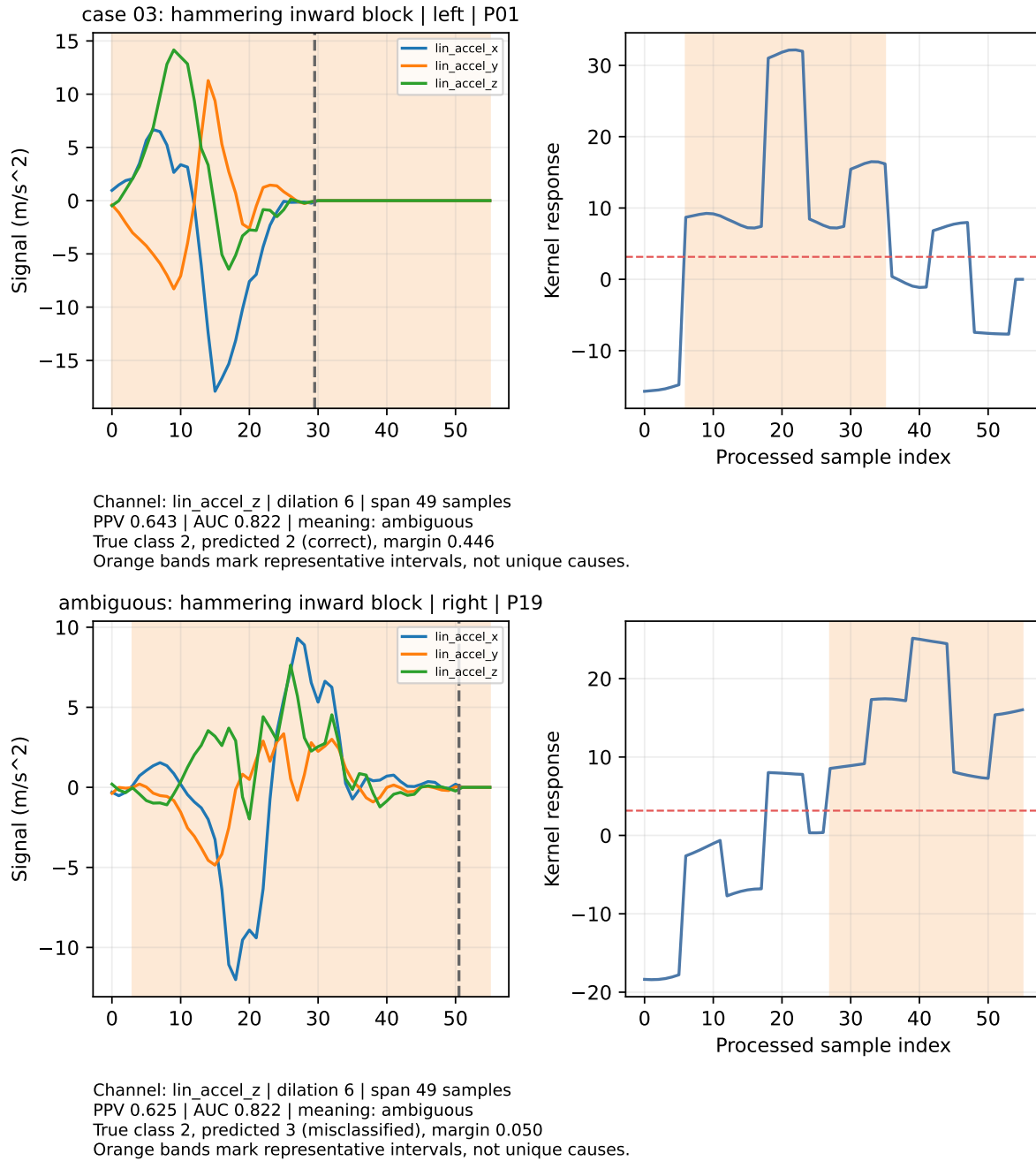


Figure 7: Task 1.3 ambiguous and failure-pattern checks. These cases keep the interpretation limits visible, especially when padding or misclassification affects the pattern.

Figure 8 checks whether the selected features separate classes in the held-out data and summarizes the four case studies. The boxplots support the selected patterns, but they still do not prove causation. The summary table makes the mixed result explicit: two cases are plausible and two are ambiguous. I treat that as useful because explainability should also show where the model evidence is hard to interpret.

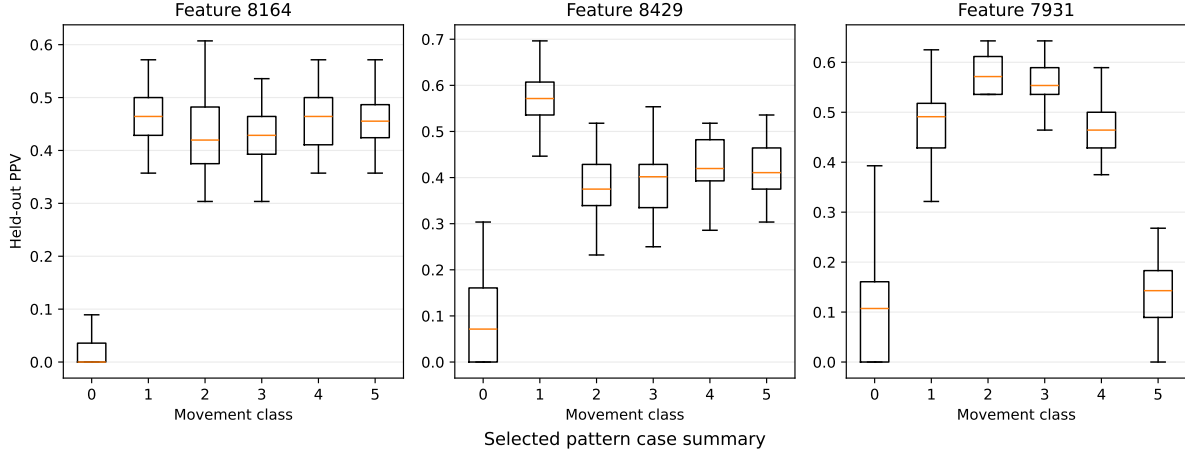


Figure 8: Task 1.3 distribution and case summary. Feature distributions support the selected cases, while the table keeps the human-meaningfulness labels explicit.

My final interpretation is that XROCKET can help a human reviewer see which parts of the sensor signal the model is using. That could support a future human-in-the-loop learning tool, because an instructor or researcher could inspect highlighted regions instead of only seeing a class label. However, this project does not validate coaching feedback, expertise assessment, force estimation, or learning gains. Those would require extra ground truth and separate evaluation.

Robustness, Confounds, And Problems Encountered

In the leakage checks, I did not find signs that the reported results were being driven by participant leakage or simple label leakage. Label-shuffle controls stayed low: the strongest mean macro F1 was 0.1605 and the maximum fold/seed macro F1 was 0.3151. Metadata-only controls using sequence length, padding fraction, and arm code also stayed low, with the strongest mean macro F1 0.2158.

The main problems encountered were:

- raw CSVs had no timestamps, so realized sampling rate and jitter are unknown;
- sequence lengths varied, requiring right-padding to 56 samples;
- XROCKET has no valid-timestep mask, so padding affects some predictions and temporal explanations;
- feature metadata required a local adapter and tests because flattened feature order had to be aligned with saved feature columns;
- channel-level rankings were less stable than family, axis, and temporal-scale rankings;
- labels 2 and 3 were the weakest classes and are mutually confused.

Limitations And Threats To Validity

The dataset is small: 240 samples from 20 participants. Results are descriptive and assignment-focused, not a clinical, coaching, diagnostic, or injury-prevention validation. Participant-independent folds reduce leakage, but they do not prove generalization beyond the KSAS protocol.

Axis interpretation is limited by phone placement and device coordinates. The phone was intended to be worn along the forearm, but participant-specific placement and orientation consistency were not available. Arm-stratified checks found that the top pooled axis differs by arm code, so pooled z-axis evidence must be read with an arm-orientation caveat.

Temporal interpretation is limited by missing timestamps and padding. Receptive fields are reported in samples first, with approximate nominal-50-Hz seconds only as context. Pattern interpretations are heuristic and must not be treated as validated biomechanics.

Reproducibility And Repository

Repository URL: <https://github.com/Forest904/ksas-xrocket-hmc.git>

Submission identifier: Git tag `v1.0-submission` on branch `main`.

The canonical Python environment is `pyproject.toml` plus `uv.lock` targeting Python 3.11. Raw KSAS data must be placed locally under `data/raw/KSAS-Dataset/`; raw participant CSVs are ignored by Git.

The compact final checks are:

```
uv sync --locked
make reproduce
make figures
make report
pdftoppm -png reports/ksas_xrocket_hmc_report.pdf tmp/pdfs/ksas_report
```

The full artifact rebuild sequence is documented in `docs/reproducibility.md`. Some commands require `--overwrite` if rerunning over existing result directories. The committed `results/` directory contains the final artifacts used by this report.

No new repository license is added in this submission. Original project code is provided for academic assignment review, raw KSAS data are excluded, and XROCKET is attributed as pinned course-authorized upstream software without a public upstream license file.

Generative-AI disclosure

I used Codex to support software setup, debugging, tests, report drafting, and language editing. I reviewed and verified the retained code, results, and text.

Conclusion

This project produced a reproducible XROCKET analysis and final PDF report for the KSAS inertial dataset. The main scientific conclusion is cautious: the model uses gravity and device-frame orientation/acceleration structure, relies most strongly on long spans in the padded XROCKET representation, and exposes representative pattern intervals that can guide expert review. The evidence is useful for explainable human motion computing, but it remains sensor-coordinate and protocol-specific rather than validated biomechanical or pedagogical truth.

References

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